

Telco Customer Churn Prediction

Business Report

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Executive Summary

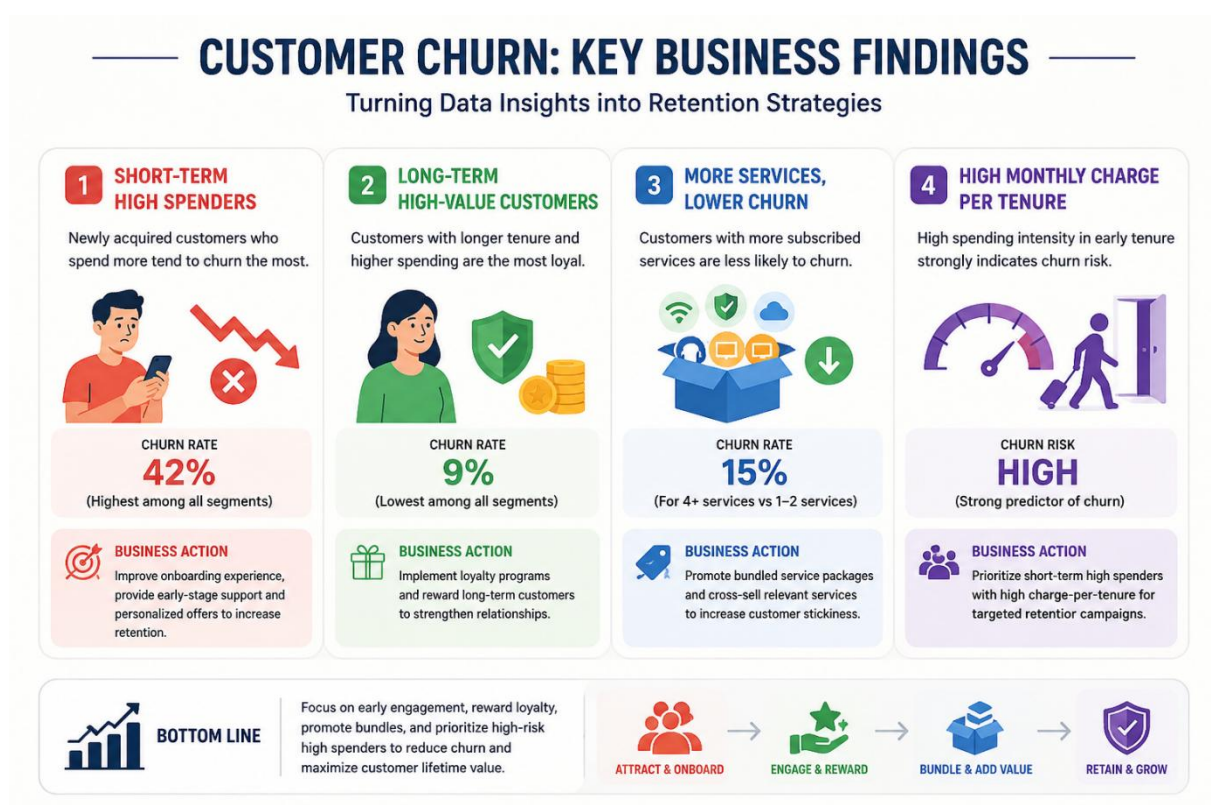
Customer churn remains one of the most significant challenges facing telecommunications providers, as acquiring new customers is generally more expensive than retaining existing ones. This analysis was conducted to identify customer segments most at risk of churn and to develop actionable retention strategies.

Using customer demographic information, service subscriptions, billing behavior, and contract information, several behavioral patterns associated with churn were identified. The analysis revealed that customer lifecycle stage, service adoption, and spending behavior are strongly associated with customer retention outcomes.

Four key business insights emerged from the analysis:

- Short-term high-spending customers exhibited elevated churn rates.
- Long-term high-value customers demonstrated strong retention behavior.
- Customers subscribed to more services were less likely to churn.
- Spending intensity relative to customer tenure emerged as a strong indicator of churn risk.

These findings provide opportunities for targeted customer retention initiatives, improved onboarding experiences, and more effective customer loyalty programs



Business Problem

Customer churn directly impacts revenue, customer lifetime value, and customer acquisition costs. Understanding which customers are most likely to leave enables organizations to intervene proactively before churn occurs.

The objectives of this analysis were to:

- Identify customer characteristics associated with churn.
- Understand behavioral patterns across different customer segments.
- Develop actionable recommendations to improve customer retention.
- Support future implementation of predictive churn monitoring systems.

Key Findings

1. Short-Term High Spenders

Finding

Customers with below-median tenure and above-median monthly charges demonstrated substantially higher churn rates than the overall customer population.

These customers generate strong revenue early in the customer lifecycle but are also more likely to leave before becoming established long-term customers.

Business Interpretation

This segment may have:

- Higher service expectations.
- Greater sensitivity to pricing.
- Lower attachment to the company.
- Increased likelihood of evaluating competing providers.

Recommendation

Develop targeted onboarding and retention programs for newly acquired premium customers, including:

- Enhanced onboarding support.
- Personalized service recommendations.
- Early retention incentives.
- Proactive customer success outreach.

2. Long-Term High-Value Customers

Finding

Customers with above-median tenure and above-median monthly charges demonstrated significantly lower churn rates.

These customers represent a highly valuable segment due to their combination of long customer relationships and strong revenue contribution.

Business Interpretation

Long-term premium customers appear to have established trust in the provider and may benefit from additional recognition and rewards.

Recommendation

Introduce customer loyalty initiatives such as:

- Loyalty reward programs.
- Exclusive promotions.
- Premium customer benefits.
- Early access to new services.

Maintaining retention within this segment should remain a strategic priority.

3. Service Bundling and Ecosystem Engagement

Finding

Customers subscribed to a greater number of telecom services consistently demonstrated lower churn rates.

As service adoption increased, churn rates generally decreased.

Business Interpretation

Customers integrated into multiple services are likely to experience higher switching costs and greater dependency on the provider's ecosystem.

Recommendation

Increase adoption of bundled service packages through:

- Cross-selling initiatives.
- Bundle discounts.
- Service upgrade campaigns.
- Personalized product recommendations.

Encouraging customers to adopt additional services may improve long-term retention.

4. Spending Intensity and Early Churn Risk

Finding

The engineered metric `MonthlyChargePerTenure` emerged as one of the strongest indicators of churn risk.

Customers exhibiting high spending relative to their customer lifetime were significantly more likely to churn.

Business Interpretation

This finding suggests that high-spending customers who have not yet established a long-term relationship with the company require additional attention during the early stages of their customer journey.

Recommendation

Use spending intensity as an early warning indicator for retention programs.

Potential actions include:

- Automated churn-risk monitoring.
- Proactive customer outreach.
- Targeted retention campaigns.
- Personalized pricing and service offers.

Recommended Retention Strategy

Immediate Actions (0–6 Months)

- Improve onboarding experiences for premium customers.
- Identify high-risk short-term customers earlier.
- Launch targeted retention campaigns for newly acquired high spenders.

Medium-Term Actions (6–12 Months)

- Increase cross-selling and service bundle adoption.
- Promote bundled offerings to existing customers.
- Encourage ecosystem engagement across multiple services.

Long-Term Actions (12+ Months)

- Establish customer loyalty programs.
- Develop premium customer retention initiatives.
- Create long-term rewards for high-value customers.

Expected Business Impact

Implementing these recommendations may contribute to:

- Improved customer retention.
- Increased customer lifetime value.
- Reduced customer acquisition costs.
- Higher customer satisfaction.
- More effective allocation of retention resources.

Conclusion

This analysis demonstrates that customer churn is influenced by more than demographic characteristics alone. Customer lifecycle stage, service adoption, and spending behavior emerged as critical factors associated with retention outcomes.

By focusing retention efforts on short-term high spenders, encouraging broader service adoption, and strengthening loyalty among long-term high-value customers, telecommunications providers can better manage churn risk and improve long-term business performance.

The findings from this project provide a foundation for both targeted retention strategies and future predictive churn monitoring systems.